Lec-2

02/06/26 10:23 AM

Ques 21:
n = 5

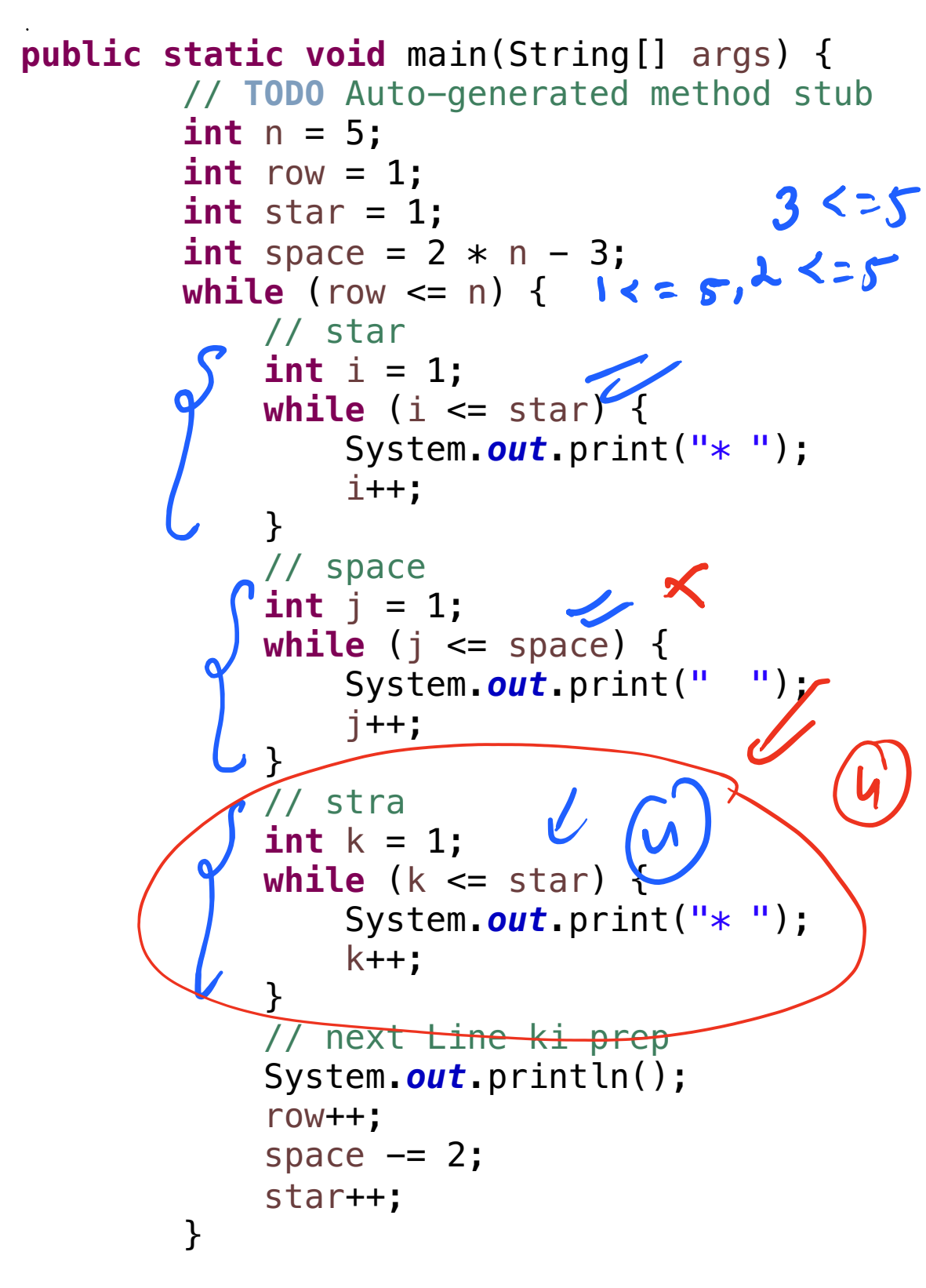
```
*                 *
*  *           *  *
*  *  *     *  *  *
*  *  *  *  *  *  *
*  *  *  *  *  *  *  *
```

2xn-1-2
= 2xN-3

2xn-1

N = 5 Row = 1 Star = 1 Space = 2xN-3

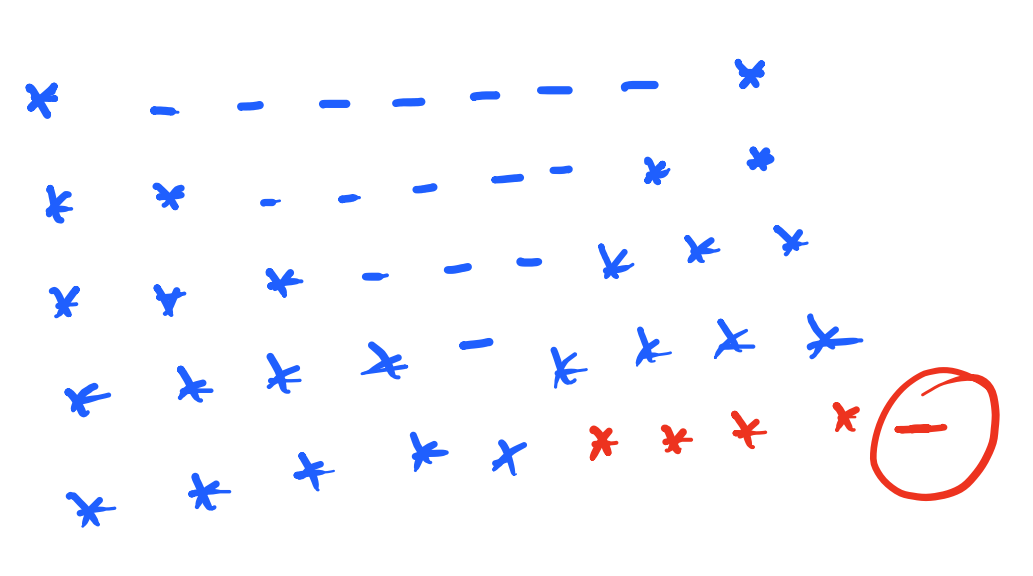
while (Row ≤ n) {
// star – i ≤ star
// Space j ≤ space
// star k ≤ star
// next line prep
Row++
S y So()
Star++
Space -= 2
}

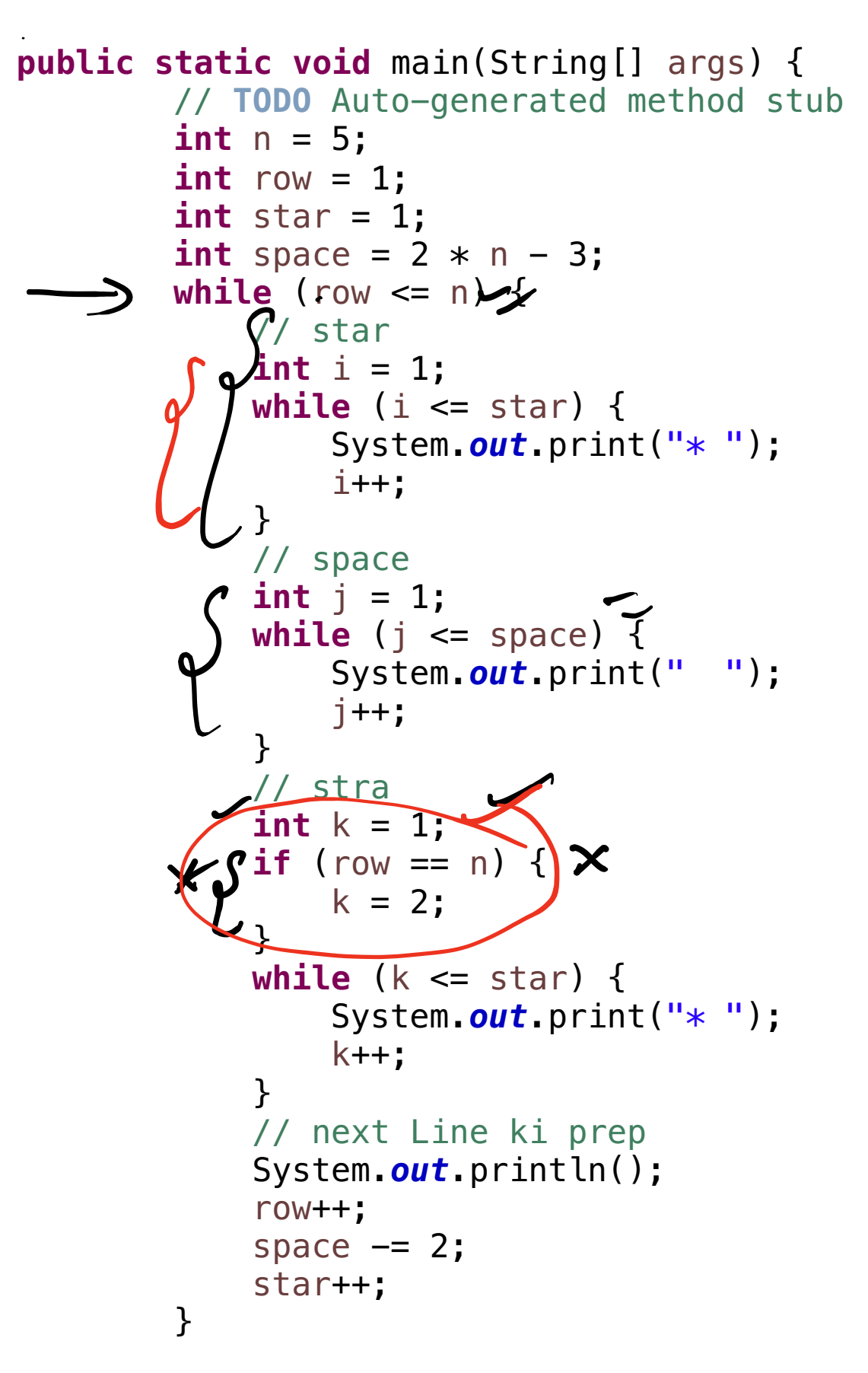


```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = 2 * n - 3;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // space
        int j = 1;
        while (j <= space) {
            System.out.print("  ");
            j++;
        }
        // stra
        int k = 1;
        while (k <= star) {
            System.out.print("* ");
            k++;
        }
        // next Line ki prep
        System.out.println();
        row++;
        space -= 2;
        star++;
    }
```

3 <= 5 4 <= 5 1 <= 5, 2 <= 5 4

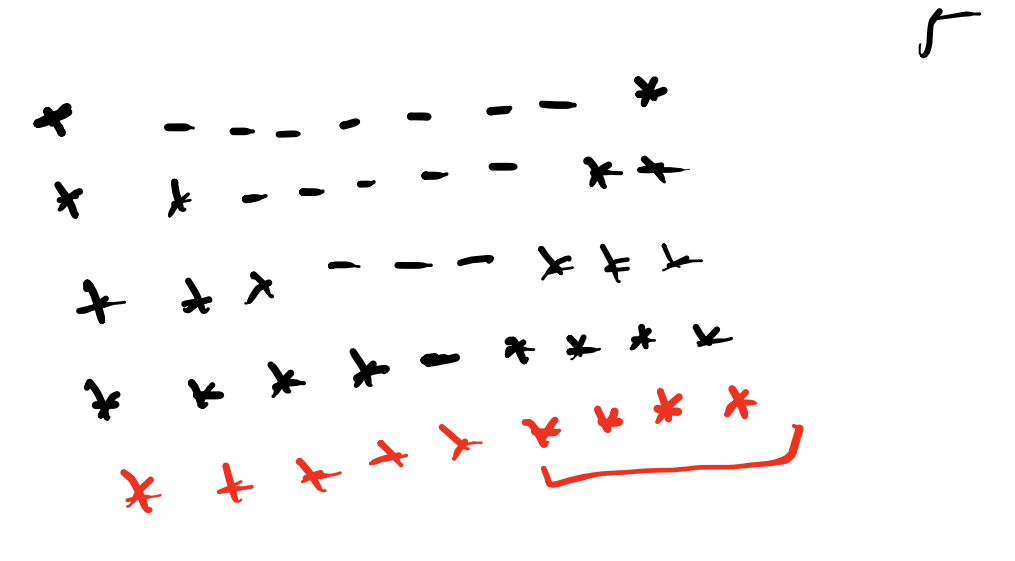
N = 5 Row = ~~1~~ ~~2~~ ~~3~~ ⑤ 4 Star = ~~1~~ ~~2~~ 3 5 4 Space = ~~7~~ 5 3 1 -1





```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = 2 * n - 3;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // space
        int j = 1;
        while (j <= space) {
            System.out.print("  ");
            j++;
        }
        // stra
        int k = 1;
        if (row == n) {
            k = 2;
        }
        while (k <= star) {
            System.out.print("* ");
            k++;
        }
        // next Line ki prep
        System.out.println();
        row++;
        space -= 2;
        star++;
    }
```

N = 5 Row = ~~1~~ ~~2~~ 3 4 5 Star = ~~1~~ 2 3 4 5 Space = ~~7~~ 5 3 1 -1



Ques 22:
n = 5

```
*  *  *  *  *  *  *  *  *
*  *  *  *     *  *  *  *
*  *  *           *  *  *
*  *                 *  *
*                       *
```

Row = 1 Star = N Space = -1
while (Row ≤ n) {
// Star
// Space
// Star
next Line Pre
Star--
Row++
S+So()
Space += 2
}

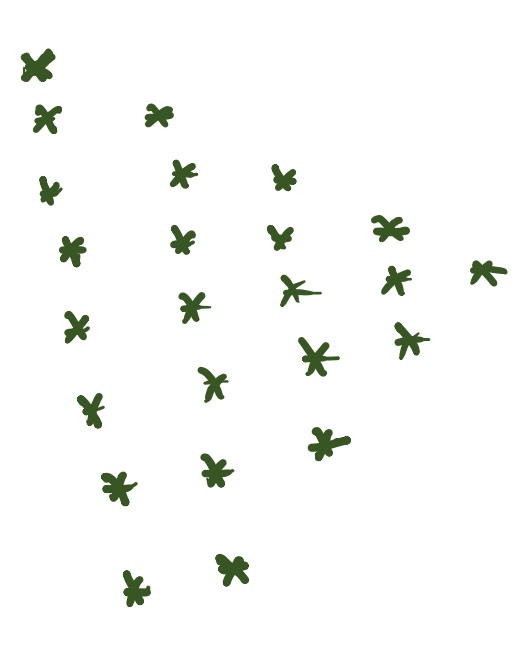
Ques 13:
n = 5

```
*
*  *
*  *  *
*  *  *  *
*  *  *  *  *
*  *  *  *
*  *  *
*  *
*
```

Row = 1, 2, 3, 4
Star++
Row = 5, 6, 7, 8
Star--

Row = ~~1~~ ~~2~~ 3 4 5 ... 8 9 Star = ~~1~~ 2 3 4 ⑤ 4 3 2 1

while (Row ≤ 2xn-1) {
int i = 1
while (i ≤ star) {
SYSO(*)
i++
}
// mirror
if (Row < n) {
Star++
} else {
Star--
}
// next prep
SYSO()
Row++
}



Ques 15:
n = 5

```
*  *  *  *  *
      *  *  *  *
         *  *  *
            *  *
               *
            *  *
         *  *  *
      *  *  *  *
*  *  *  *  *
```

Row = 1 Space = 0 Star = n
while (Row ≤ 2xn-1) {
// Space
// Star
mirror
if (Row < n) {
Star--
Space += 2
} else {
Star++
Space -= 2
}
Row++
S+So()
}

N = 7

```
*  *  *     *  *  *
*  *           *  *
*                 *

*                 *
*  *           *  *
*  *  *     *  *  *
```

7/2 = 3   7/2 = 3.5

Row = 1 Star = N/2 Space = 1
while (Row ≤ n) {
// Star
// Space
// Star
// mirror
if (Row < N/2 + 1) {
Star--, Space += 2
} else {
Star++, Space -= 2
}
Row++
S+So()
}

Ques 19:
n = 7

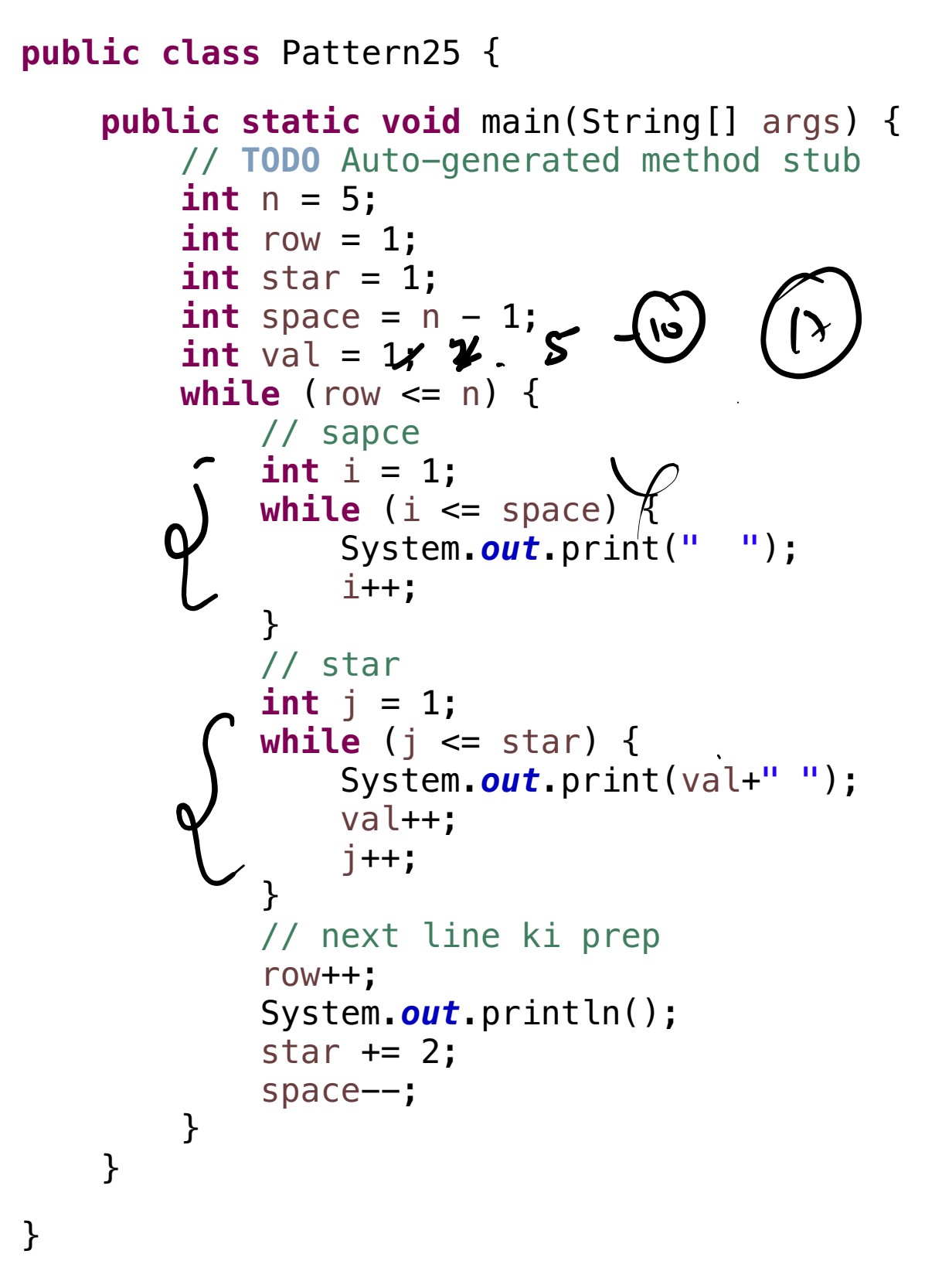
```
*  *  *  *  *  *  *
*  *  *     *  *  *
*  *           *  *
*                 *
*  *           *  *
*  *  *     *  *  *
*  *  *  *  *  *  *
```

Row = 1 Star = N/2 + 1 Space = -1
while (Row ≤ n) {
// Star
// Space
// Star
mirror
if (Row < N/2 + 1) {
Star--, Space += 2
} else {
Star++
Space -= 2
}
SYSO()
Row++
}

Star

Ques 25:
n = 5

```
            1
         2  3  4
      5  6  7  8  9
  10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```



```
public class Pattern25 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        int val = 1;
        while (row <= n) {
            // sapce
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print(val+" ");
                val++;
                j++;
            }
            // next line ki prep
            row++;
            System.out.println();
            star += 2;
            space--;
        }
    }
}
```

val = ~~1~~ ~~2~~ 5 ⑩ ⑰

N = 5 Row = ~~1~~ 2 3 4 5 Star = ~~1~~ 3 5 7 9 Space = 4 3 2 1 0

- - - - 1
- - - 2 3 4
- - 5 6 7 8 9
- 10 11 12 13 14 15 16
17
㉑

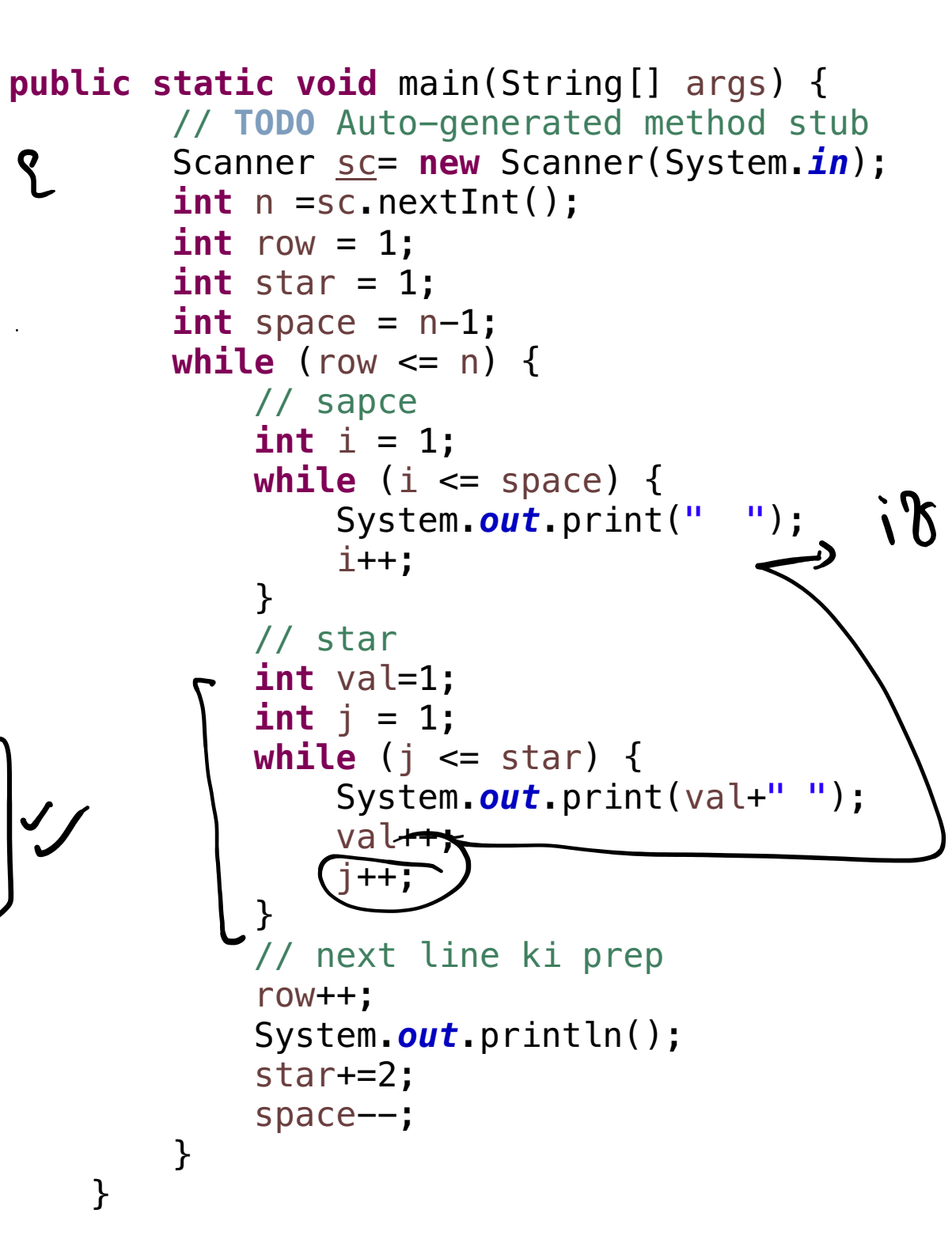
Ques 27:
n = 5

```
            1
         1  2  1
      1  2  3  2  1
   1  2  3  4  3  2  1
1  2  3  4  5  4  3  2  1
```

① 3 5 7 9

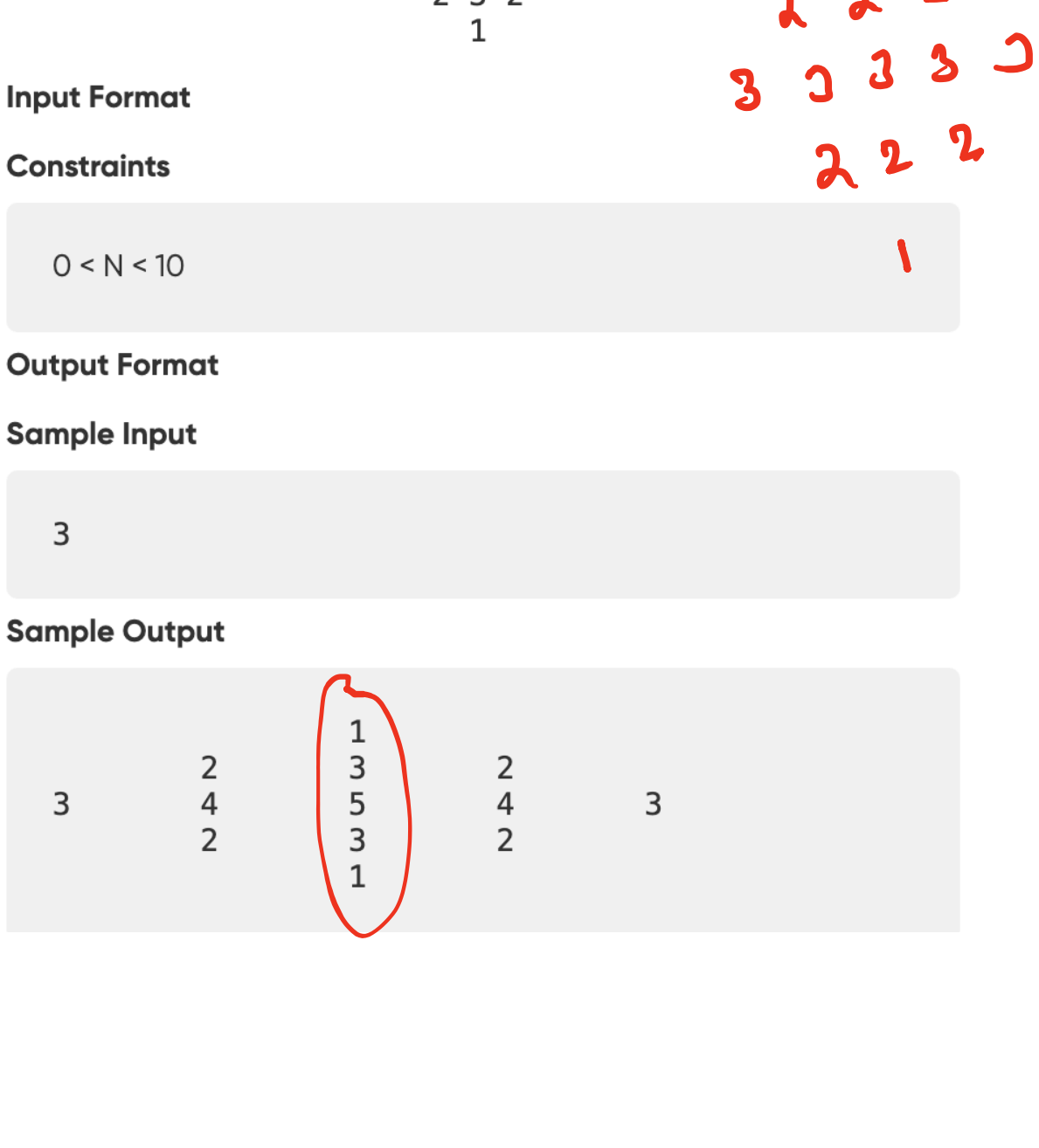
star/2 + 1

if ( ) {
} else {
}



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    Scanner sc= new Scanner(System.in);
    int n =sc.nextInt();
    int row = 1;
    int star = 1;
    int space = n-1;
    while (row <= n) {
        // sapce
        int i = 1;
        while (i <= space) {
            System.out.print("  ");
            i++;
        }
        // star
        int val=1;
        int j = 1;
        while (j <= star) {
            System.out.print(val+" ");
            val++;
            j++;
        }
        // next line ki prep
        row++;
        System.out.println();
        star+=2;
        space--;
    }
}
```

if (j < star/2 + 1) {
val++
} else {
val--
}



Take N (number of rows), print the following pattern (for N = 3).

```
    1
  2 3 2
3 4 5 4 3
  2 3 2
    1
```

Input Format

Constraints

0 < N < 10

Output Format

Sample Input

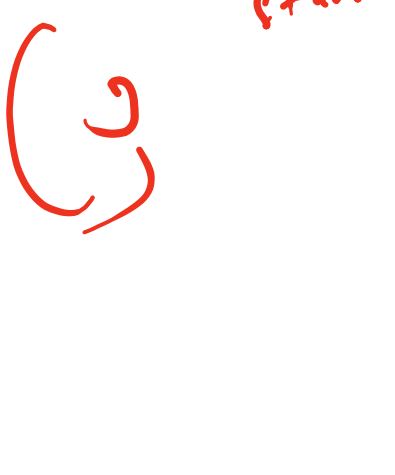
3

Sample Output

```
3    2    1    2    3
     4    3    4
     2    5    2
          3
          1
```

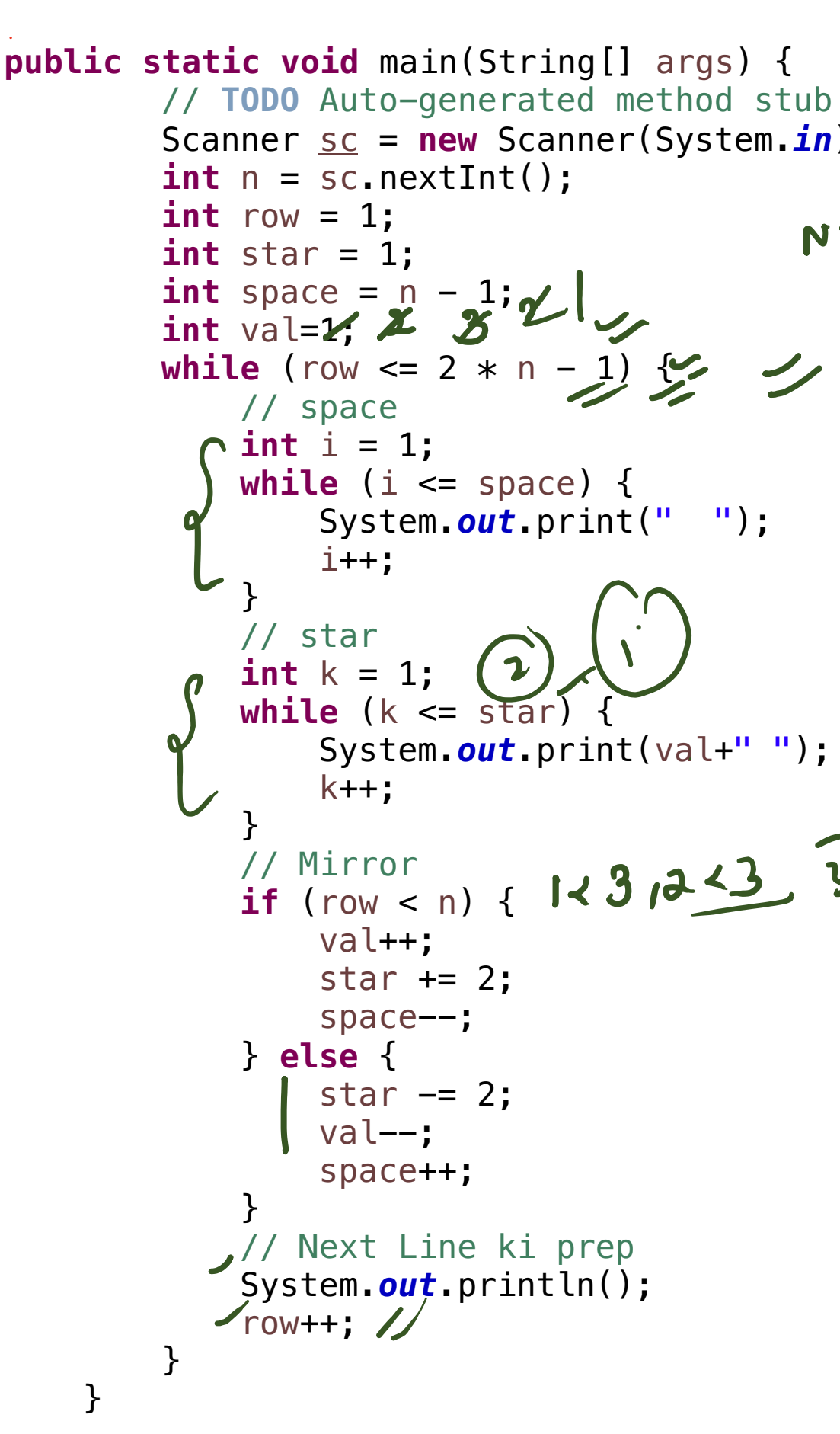
1
2 2 2
3 3 3 3 3
2 2 2
1

①



Row = 1
Star = 1
Space = n-1

while (Row ≤ 2xn-1) {
// Space
// Star
if (Row < n) {
Space
Star += 2
}



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    int row = 1;
    int star = 1;
    int space = n - 1;
    int val=1;
    while (row <= 2 * n - 1) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print("  ");
            i++;
        }
        // star
        int k = 1;
        while (k <= star) {
            System.out.print(val+" ");
            k++;
        }
        // Mirror
        if (row < n) {
            val++;
            star += 2;
            space--;
        } else {
            star -= 2;
            val--;
            space++;
        }
        // Next Line ki prep
        System.out.println();
        row++;
    }
}
```

val = ~~1~~ 2 3 2 1

1<3 2<3 3<3

N = 3

Row = ~~1~~ ~~2~~ 3 4 5 Star = ~~1~~ ~~3~~ 5 3 1 Space = ~~2~~ 1 0 1 2

- - 1
- 2 2 2
3 3 3 3 3
- 2 2 2
- - 1